

Chapter 6: Pickup-and-Delivery Problems for Goods Transportation

Vehicle Routing: Problems, Methods, and Applications
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6.1 Introduction: What Are Pickup-and-Delivery Problems? I

Pickup-and-Delivery Problems (PDPs) constitute an important and broad family of routing problems in which goods or passengers must be transported between origins and destinations. Unlike the classical Vehicle Routing Problem (VRP) – in which goods flow only *from* a depot *to* customers – a PDP involves two-sided requests: for each request there is a **pickup location** where the item is collected and a **delivery location** where the item is dropped off.

Formal Definition of a PDP Request

A **request** r specifies:

- a **pickup node** p_r : the origin where the item must be collected,
- a **delivery node** d_r : the destination where the item must be delivered,
- a **load** q_r : the quantity (weight, volume) associated with the request,
- (optionally) **time windows** $[e_{p_r}, l_{p_r}]$ and $[e_{d_r}, l_{d_r}]$ specifying the earliest and latest service times at both endpoints.

The fundamental constraint is that the pickup must occur *before* the delivery on the same vehicle's route.

6.1 Introduction: What Are Pickup-and-Delivery Problems? II

Three structural categories. PDPs are classified according to the relationship between origins and destinations of requests:

- ① **Many-to-Many (M-M):** Any node can be both an origin and a destination. This arises in transportation networks such as freight exchange platforms, where goods move between arbitrary pairs of nodes. Dial-a-ride problems without fixed depot constraints are special cases.
- ② **One-to-Many-to-One (1-M-1):** Goods originate from a *single depot*, are delivered to customers, and empty or loaded vehicles return to the same depot. This models distribution networks where a vehicle may also collect goods (e.g., returned merchandise) during the same tour.
- ③ **One-to-One (1-1):** Each request specifies a *unique* pickup location and a *unique* delivery location. Items cannot be transhipped: the same vehicle that picks up an item must deliver it. This is the most constrained and the most widely studied variant for goods transportation.

6.1 Introduction: What Are Pickup-and-Delivery Problems? III

Three Categories of Pickup-and-Delivery Problems

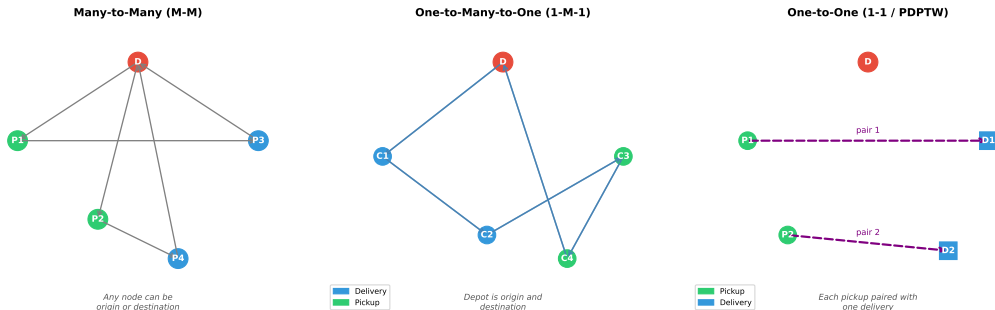


Figure: The three structural categories of PDPs. Green = pickup nodes, blue = delivery nodes, red = depot. In M-M any node is both; in 1-M-1 the depot is origin/destination; in 1-1 each pickup is paired with exactly one delivery.

6.1 Introduction: What Are Pickup-and-Delivery Problems? IV

Additional problem dimensions. Beyond the structural category, PDPs are further characterised by:

- **Fleet composition:** homogeneous (all vehicles identical) or heterogeneous (vehicles differ in capacity, cost, or type).
- **Time windows:** hard (service outside the window is forbidden) or soft (violations are penalised in the objective).
- **Loading constraints:** LIFO (Last-In First-Out), FIFO (First-In First-Out), or full three-dimensional packing feasibility.
- **Objective:** minimise total travel distance or time, minimise number of vehicles, or a weighted combination.

6.1 Introduction: What Are Pickup-and-Delivery Problems? V

Key Insight: Why PDPs Are Harder Than VRPs

In a standard VRP, feasibility checking for a route requires only that the total demand does not exceed capacity. In a PDP, feasibility also requires: (1) the pickup precedes its paired delivery (*precedence* constraint), (2) the vehicle load never exceeds capacity at any point on the route (*capacity* constraint along the tour, not just in total), and (3) all time windows are satisfied. The combination of these constraints makes PDPs significantly harder combinatorially.

Dynamic and stochastic variants. Many real-world applications involve *dynamic* PDPs in which requests arrive in real time (courier services, taxi dispatching), or *stochastic* PDPs in which demand or travel times are uncertain. This chapter focuses on the static, deterministic setting.

6.2 Many-to-Many (M-M) Problems I

In the Many-to-Many PDP, multiple origins and multiple destinations exist, and any node can serve as both an origin and a destination. The classic example is a transportation network where goods are to be moved between arbitrary pairs of nodes, typically using a fleet of vehicles.

M-M PDP: Mathematical Formulation

Let $G = (V, A)$ be a directed graph where $V = \{0, 1, \dots, n\}$ is the node set and A is the arc set. Each node $i \in V$ has a **supply** s_i (positive if it produces goods, negative if it consumes goods). We seek routes for m vehicles of capacity Q such that:

$$\min \sum_{k=1}^m \sum_{(i,j) \in A} c_{ij} x_{ij}^k$$

subject to flow conservation at every node, vehicle capacity constraints, and the requirement that all supply-demand imbalances are resolved. Here c_{ij} is the travel cost of arc (i, j) and $x_{ij}^k \in \{0, 1\}$ indicates whether vehicle k traverses arc (i, j) .

6.2 Many-to-Many (M-M) Problems II

Dial-a-Ride relaxation. A well-known special case of M-M problems is the **Dial-a-Ride Problem (DARP)**, in which users specify an origin and destination and a vehicle must transport them. In the goods context, the “passengers” are replaced by freight units. When the depot is not a fixed constraint – meaning vehicles may start and end at any node – the problem becomes a pure M-M problem.

6.2 Many-to-Many (M-M) Problems III

Single-vehicle case. Hernandez-Perez and Salazar-Gonzalez (2004) introduced the **1-PDTSP** (one-vehicle Pickup-and-Delivery Travelling Salesman Problem) for M-M settings. A complete graph $G = (V, A)$ is given; each node i has a supply s_i (positive = source, negative = sink), and a single vehicle of capacity Q must visit all nodes. The 1-PDTSP can be formulated as:

6.2 Many-to-Many (M-M) Problems IV

1-PDTSP Formulation

Decision variables:

- $x_{ij} \in \{0, 1\}$: equals 1 if arc (i, j) is traversed by the vehicle.
- $y_i \geq 0$: vehicle load upon departure from node i .

Objective:

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij}$$

Constraints:

$$\sum_j x_{ij} = 1 \quad \forall i \in V \quad (1: \text{ each node left once})$$

$$\sum_i x_{ij} = 1 \quad \forall j \in V \quad (2: \text{ each node entered once})$$

$$y_j \geq y_i + s_j \cdot x_{ij} - Q(1 - x_{ij}) \quad \forall i \neq j \quad (3: \text{ load tracking})$$

$$0 \leq y_i \leq Q \quad \forall i \in V \quad (4: \text{ capacity})$$

$$x_{ij} \in \{0, 1\} \quad (5: \text{ binary})$$

6.3 One-to-Many-to-One (1-M-1) Problems I

In the One-to-Many-to-One PDP, all goods originate from (or return to) a single depot. Vehicles depart from the depot carrying goods for delivery, visit customers to make deliveries, and may also collect goods (pickups) from customers during the same tour before returning to the depot. This structure models real-world applications such as beverage distribution (delivering full crates and collecting empty ones), or the simultaneous distribution and collection of products in retail networks.

6.3 One-to-Many-to-One (1-M-1) Problems II

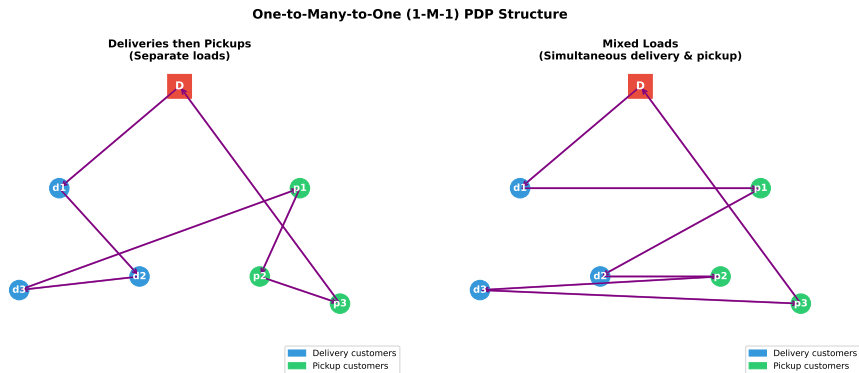


Figure: Two variants of the 1-M-1 structure. Left: deliveries precede pickups (vehicles first unload all goods, then collect). Right: mixed loads (delivery and pickup customers are interleaved on the route).

6.3 One-to-Many-to-One (1-M-1) Problems III

Two sub-cases of 1-M-1. The 1-M-1 structure can be further divided:

- ① **Deliveries then pickups:** the vehicle visits all delivery customers before visiting any pickup customer. This is a natural operational constraint when delivery and pickup goods cannot be mixed (e.g., fragile items).
- ② **Mixed loads:** delivery and pickup customers are interleaved on the route. The vehicle load at any point is the original delivery load minus goods delivered plus goods collected up to that point. The capacity constraint must hold throughout the route.

6.3 One-to-Many-to-One (1-M-1) Problems IV

The Vehicle Routing Problem with Simultaneous Pickup and Delivery (VRPSPD)

The VRPSPD is the most general form of 1-M-1. A fleet of m homogeneous vehicles, each of capacity Q , must serve n customers. Each customer i has a **delivery demand** $d_i > 0$ and a **pickup demand** $p_i \geq 0$. The vehicle must carry both when visiting customer i .

Decision variables:

- $x_{ij}^k \in \{0, 1\}$: 1 if vehicle k travels from i to j .
- $y_{ij}^k \geq 0$: delivery load on vehicle k when traversing arc (i, j) .
- $z_{ij}^k \geq 0$: pickup load on vehicle k when traversing arc (i, j) .

Objective – minimise total travel cost:

$$\min \sum_{k=1}^m \sum_{(i,j) \in A} c_{ij} x_{ij}^k$$

Core constraints: flow conservation, each customer visited exactly once, and at every arc the combined delivery and pickup loads must satisfy:

$$k \leq k \leq Q - k \quad \forall i, j$$

6.4.1 The PDPTW: Mathematical Formulation I

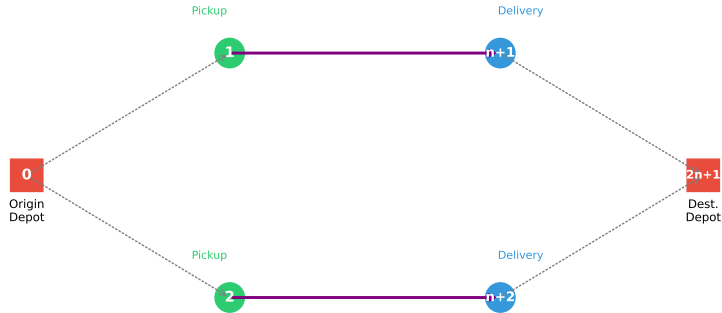
The **Pickup-and-Delivery Problem with Time Windows (PDPTW)** is the most extensively studied variant of the 1-1 PDP. In this problem, each request specifies a unique pickup location and a unique delivery location with associated time windows, and the same vehicle that picks up the goods must deliver them without intermediate transshipment.

Problem data.

- n : number of requests.
- Node set: $N = \{0, 1, \dots, 2n, 2n + 1\}$ where node 0 is the origin depot, node $2n + 1$ is the destination depot (a copy of node 0), nodes $1, \dots, n$ are pickup nodes, and nodes $n + 1, \dots, 2n$ are delivery nodes.
- Each request $i \in \{1, \dots, n\}$ consists of pickup node i and delivery node $n + i$.
- c_{ij} : travel cost from node i to node j .
- t_{ij} : travel time from node i to node j .
- $[e_i, l_i]$: time window at node i (service must begin in this interval).
- q_i : load change at node i ($q_i > 0$ for pickup nodes, $q_i < 0$ for delivery nodes).
- Q : vehicle capacity (maximum load at any point in the route).

6.4.1 The PDPTW: Mathematical Formulation II

PDP Graph: Single-Vehicle Request Graph
(Pickup nodes 1-n, Delivery nodes $n+1$ to $2n$, Depot 0 and $2n+1$)



Purple arrows = precedence: pickup must happen before its paired delivery

6.4.1 The PDPTW: Mathematical Formulation III

Figure: The request graph for the PDPTW. Pickup nodes (green) are numbered $1 \dots n$; delivery nodes (blue) are $n + 1 \dots 2n$; origin and destination depots are shown in red. Purple arrows indicate precedence: pickup i must precede delivery $n + i$ on the same route.

6.4.1 The PDPTW: Mathematical Formulation IV

PDPTW: Complete Integer Programme (Savelsbergh & Sol, 1995)

Decision variables:

- $x_{ij}^k \in \{0, 1\}$: equals 1 if vehicle k travels directly from node i to node j .
- $B_i^k \geq 0$: the time at which vehicle k begins service at node i .
- $L_i^k \geq 0$: the load of vehicle k when it departs node i .

Objective – minimise total travel cost:

$$\min \sum_{k \in K} \sum_{i \in N} \sum_{j \in N} c_{ij} x_{ij}^k \quad (\text{PDPTW-OBJ})$$

Constraints:

$$\sum_{k \in K} \sum_{j \in N} x_{ij}^k = 1 \quad \forall i \in P \quad (\text{C1}) \quad (1)$$

$$\sum_{j \in N} x_{0j}^k = 1 \quad \forall k \in K \quad (\text{C2}) \quad (2)$$

$$\sum_{i: 0 \leq i \leq n-1} x_{i,n}^k = 1 \quad \forall k \in K \quad (\text{C3}) \quad (3)$$

6.4.2 Exact Methods for PDPTW I

Exact methods for the PDPTW guarantee an optimal solution at the cost of potentially exponential running time. They are practical for instances up to roughly 50–75 requests depending on the tightness of the time windows.

Branch-and-cut (B&C). The Branch-and-Cut method solves a sequence of LP relaxations strengthened by cutting planes, combined with a branch-and-bound tree search. The general scheme is:

- 1 **Solve LP relaxation** at the root node. If the optimal LP solution is integer, stop – it is the optimal PDPTW solution. Otherwise, proceed.
- 2 **Add cuts:** separate violated subtour elimination constraints, capacity cuts, and PDP-specific precedence cuts to tighten the LP relaxation.
- 3 **Branch:** select a fractional variable x_{ij}^k and create two sub-problems (branches): one fixing $x_{ij}^k = 0$ and one fixing $x_{ij}^k = 1$.
- 4 **Prune:** if the LP lower bound of a node exceeds the incumbent upper bound, prune that branch.
- 5 **Repeat** until all nodes are resolved.

6.4.2 Exact Methods for PDPTW II

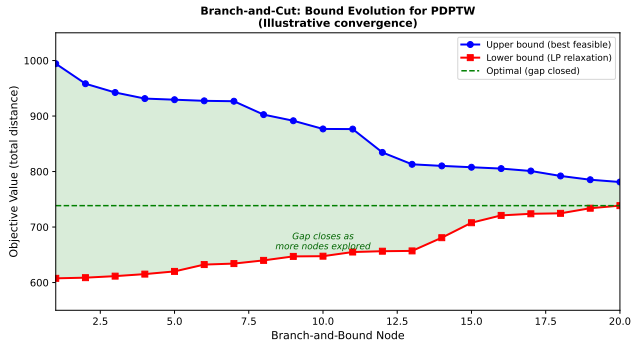


Figure: Illustrative convergence of the branch-and-cut method. The upper bound (best feasible solution found) decreases while the lower bound (LP relaxation) increases as more nodes are explored, until the gap is closed.

6.4.2 Exact Methods for PDPTW III

Key B&C results for PDPTW. Cordeau (2006) proposed a branch-and-cut algorithm for the PDPTW that solved instances with up to 75 requests. The key ingredients were:

- **k -path inequalities:** violated if the current LP solution routes more than k paths through a cut set, beyond what the capacity allows.
- **Clique-based lifted cover inequalities:** tighten capacity constraints by exploiting incompatibility between requests.
- **Branching on 3-cycles:** when many 3-cycle variables are fractional, branching on their indicator variables tends to close the LP gap quickly.

Dynamic programming for single-vehicle PDPTW. When only one vehicle is available, the problem reduces to a sequencing problem. Exact DP formulations track the current time, current load, and the set of requests whose pickup has been made but delivery not yet completed (*open requests*). The state space is exponential in the number of open requests, making DP practical only for small instances.

Column generation / Branch-and-Price. An alternative exact approach formulates the PDPTW as a **set-partitioning problem**:

$$\min \sum_{r \in \Omega} c_r \lambda_r \quad \text{s.t.} \quad \sum_{r \in \Omega} a_{ir} \lambda_r = 1 \quad \forall i, \quad \lambda_r \in \{0, 1\}$$

6.4.2 Exact Methods for PDPTW IV

where Ω is the set of all feasible routes, c_r is the cost of route r , and $a_{ir} = 1$ if route r serves request i . Since $|\Omega|$ is exponentially large, column generation solves the LP relaxation by generating only routes with negative reduced cost – the **pricing sub-problem** – which is itself a shortest-path problem with resource constraints (SPPRC).

6.4.2 Exact Methods for PDPTW V

Complexity summary for exact methods:

Method	Instance size	Key reference
Branch-and-cut	≤ 75 requests	Cordeau (2006)
Branch-and-price	≤ 50 requests	Dumas et al. (1991)
Dynamic programming	≤ 20 requests	Rais et al. (2014)

Key Takeaway: Exact Methods

Branch-and-cut is the most powerful exact method for the PDPTW and can reliably solve instances with up to 75 requests. Beyond this scale, metaheuristics are necessary. The bottleneck for exact methods is the tightness of the LP bound: tight time windows dramatically reduce the search space and allow larger instances to be solved exactly.

6.4.3 Heuristics and Metaheuristics for PDPTW I

Because the PDPTW is NP-hard, practical solvers rely on heuristics and metaheuristics that trade off solution quality for computational efficiency. The goal is to find near-optimal solutions quickly, especially for real-world instances with 100–1000 requests.

Construction heuristics. The most widely used construction approach for the PDPTW is the **parallel or sequential insertion heuristic**:

- 1 Start with m routes, each containing only the depot.
- 2 For each unassigned request, compute the **best feasible insertion** into every existing route – the position that minimises the increase in total cost subject to precedence, capacity, and time-window constraints.
- 3 Insert the request that can be cheaply inserted first (cheapest-first) or most urgently (most constrained first).
- 4 Repeat until all requests are assigned.

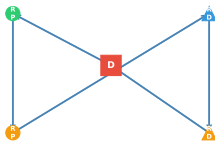
6.4.3 Heuristics and Metaheuristics for PDPTW II

Insertion Heuristic for PDPTW: Step-by-Step

Step 1: Insert Request 1
(cheapest feasible position)



Step 2: Insert Request 2
(best cost increase)



Step 3: Insert Request 3
(final route)

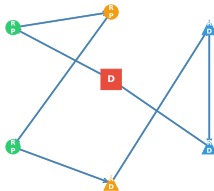


Figure: Three-step insertion heuristic for a PDPTW with 3 requests. Step 1 inserts Request 1 (orange pair) to form the first partial route. Step 2 inserts Request 2 at the best feasible position within the existing route. Step 3 inserts Request 3 (orange) to produce the final solution.

6.4.3 Heuristics and Metaheuristics for PDPTW III

Tabu Search (TS). Tabu Search was applied to the PDPTW by Nanry and Barnes (2000) and later extended by Li and Lim (2003), among others. The core idea is to explore the solution space by systematically making improving or, when no improvement exists, the least-worsening moves, while maintaining a **tabu list** of recently performed moves to avoid cycling.

6.4.3 Heuristics and Metaheuristics for PDPTW IV

Tabu Search Framework for PDPTW

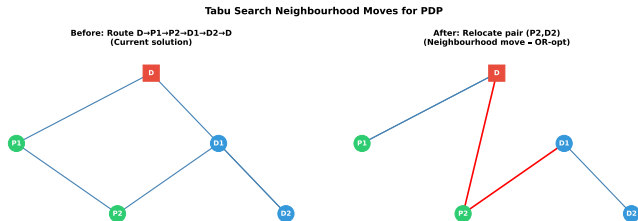
Input: an initial feasible solution s_0 ; tabu tenure T ; maximum iterations $I_{\max}$.

Output: the best solution found s^* .

- ① Set $s \leftarrow s_0$, $s^* \leftarrow s_0$, tabu list $\mathcal{T} \leftarrow \emptyset$.
- ② **For** $i = 1, 2, \dots, I_{\max}$:
 - a. Generate the neighbourhood $\mathcal{N}(s)$ of s by applying all feasible *request-relocation* moves: remove a request (p_i, d_{n+i}) from its current route and reinsert it at the best feasible position in the same or a different route.
 - b. Select the best non-tabu move m^* from $\mathcal{N}(s)$, or accept a tabu move if it satisfies the **aspiration criterion** (the resulting solution is better than s^*).
 - c. Apply m^* : $s \leftarrow s \oplus m^*$.
 - d. Add m^* to $\mathcal{T}$; remove the oldest entry if $|\mathcal{T}| > T$.
 - e. If $f(s) < f(s^*)$ then $s^* \leftarrow s$.
- ③ **Return** s^* .

6.4.3 Heuristics and Metaheuristics for PDPTW V

Neighbourhood structure for PDPTW TS. The most effective neighbourhood for PDPTW TS uses **request relocation**: remove a (pickup, delivery) pair from its current route and reinsert it into another route (or the same route at a different position). The move must maintain feasibility – precedence, capacity, and time windows.



6.4.3 Heuristics and Metaheuristics for PDPTW VI

Figure: Before and after a request-relocation neighbourhood move in PDPTW Tabu Search. The pair (P2, D2) is removed from route 1 and relocated to a different position (shown in red). After the move, both routes must remain feasible.

Why does the tabu list help? In local search without a tabu list, the algorithm tends to cycle between two solutions that are local optima of each other. The tabu list forbids reversing the most recent T moves, forcing the search to explore new regions. Setting T too small allows cycling; T too large makes the search inflexible. Adaptive tabu tenures (randomly varying between $T_{\min}$ and $T_{\max}$) are common in practice.

6.4.3 Heuristics and Metaheuristics for PDPTW VII

Large Neighbourhood Search (LNS) for PDPTW. Ropke and Pisinger (2006) proposed LNS as an effective method for the PDPTW. The algorithm iteratively:

- 1 **Destroys** part of the current solution by removing q requests (where q is a *destroy degree*, typically $q \approx 0.1 \cdot n$ to $0.4 \cdot n$).
- 2 **Repairs** the partial solution by reinserting the removed requests using a greedy insertion heuristic.
- 3 **Accepts** the new solution according to a simulated annealing criterion: s_{new} is accepted if $f(s_{\text{new}}) < f(s)$, or with probability $\exp(-(f(s_{\text{new}}) - f(s))/T)$ if $f(s_{\text{new}}) \geq f(s)$.

LNS Destroy Operators

- **Random removal:** select q requests uniformly at random.
- **Shaw removal:** remove q requests that are *similar* to each other, where similarity is measured by geographic proximity and time-window overlap. The idea is that similar requests are easier to mutually reassign.
- **Worst removal:** remove the q requests with the highest contribution to the total solution cost (the most expensive requests).

6.4.3 Heuristics and Metaheuristics for PDPTW VIII

Adaptive LNS (ALNS). Ropke and Pisinger (2006) extended LNS to the **Adaptive Large Neighbourhood Search (ALNS)** by maintaining a *pool* of destroy and repair operators, and dynamically adjusting the probability of selecting each operator based on its historical performance:

- Each operator o is assigned a score π_o that accumulates reward whenever the operator contributes to a new best solution, an improving solution, or an accepted non-improving solution.
- After every σ iterations (a *segment*), the selection probability of each operator is updated:

$$p_o \leftarrow \frac{w_o}{\sum_{o'} w_{o'}}, \quad w_o \leftarrow (1 - r)w_o + r \frac{\pi_o}{\theta_o}$$

where w_o is the weight, $r \in (0, 1)$ is a reaction factor controlling how quickly weights adapt, and θ_o is the number of times operator o was used during the segment.

- Operators that consistently find better solutions receive higher probability of being selected in subsequent iterations.

Why ALNS works. Different destroy operators are effective for different types of problem structure. Random removal provides diversification; Shaw removal provides intensification around related requests; worst removal targets high-cost areas. Adapting probabilities ensures the algorithm exploits whichever strategy is most effective for the current instance.

6.4.3 Heuristics and Metaheuristics for PDPTW IX

ALNS Numerical Illustration: Score Update

Suppose the algorithm has 3 destroy operators: Random (R), Shaw (S), Worst (W). After one segment of $\sigma = 50$ iterations, the collected statistics are:

Operator	θ_o (uses)	π_o (score)	w_o (old)	w_o (new, $r = 0.2$)
Random (R)	18	24	1.00	$0.8 \times 1.00 + 0.2 \times (24/18) = 1.067$
Shaw (S)	17	38	1.00	$0.8 \times 1.00 + 0.2 \times (38/17) = 1.247$
Worst (W)	15	12	1.00	$0.8 \times 1.00 + 0.2 \times (12/15) = 0.960$

The selection probabilities become: $p_R = 1.067/3.274 = 0.326$; $p_S = 1.247/3.274 = 0.381$; $p_W = 0.960/3.274 = 0.293$. Shaw removal, having found the most improvements, now has the highest selection probability.

6.4.3 Heuristics and Metaheuristics for PDPTW X

Branch-and-Cut integrated with heuristics. Ropke et al. (2007) combined branch-and-cut with LNS in a so-called **matheuristic** framework: the LNS is used to find a good upper bound, which is then passed to the B&C as the incumbent, greatly reducing the B&C search tree.

POPMUSIC. For very large PDPTW instances (several hundred requests), the POPMUSIC framework (Partial Optimisation Metaheuristic Under Special Intensification Conditions) by Taillard et al. has been adapted: the problem is decomposed into overlapping sub-problems of manageable size, each sub-problem is optimised by a fast metaheuristic, and the solutions are stitched back together.

6.4.3 Heuristics and Metaheuristics for PDPTW XI

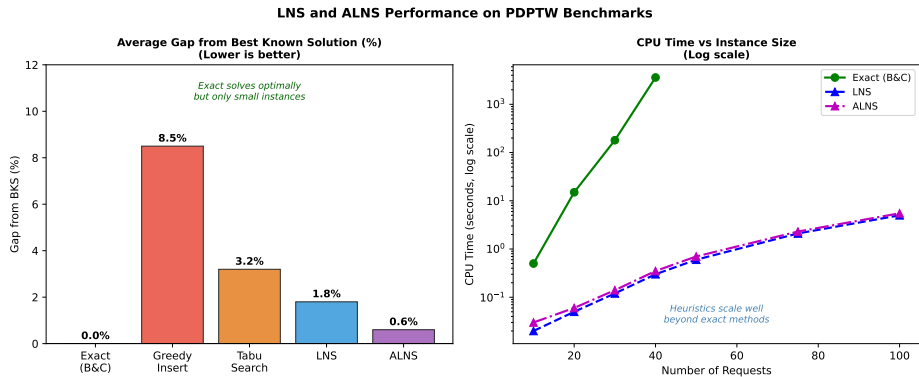


Figure: Left: average gap from best known solution for different methods on PDPTW benchmarks. ALNS achieves solutions within 0.6% of BKS. Right: CPU time vs instance size (log scale); exact B&C is practical only for small instances while LNS and ALNS scale to hundreds of requests.

6.4.3 Heuristics and Metaheuristics for PDPTW XII

Key Takeaway: Metaheuristics for PDPTW

ALNS is currently the leading metaheuristic for the PDPTW, offering the best balance of solution quality and scalability. The key design choice is the set of destroy/repair operators: combining random, similarity-based (Shaw), and worst-case removal with adaptive operator selection has proven highly effective across diverse benchmark sets.

6.4.4 Single-Vehicle PDPTW I

When the fleet consists of a single vehicle, the PDPTW reduces to the **Pickup-and-Delivery TSP with Time Windows (PDTSPW)**. This problem has been studied both as a theoretical object and as a sub-problem in column generation approaches to the multi-vehicle PDPTW.

Single-vehicle exact algorithm. Ruland and Rodin (1997) proposed a branch-and-cut algorithm specifically for the single-vehicle case. The key observation is that for a single vehicle, the constraint that the same vehicle serves both endpoints of each request is automatically satisfied. The remaining difficulty is the combination of:

- Precedence constraints: pickup i before delivery $n + i$.
- Capacity constraints: load $\in [0, Q]$ throughout.
- Time window constraints: service within $[e_i, l_i]$ at each node.

6.4.4 Single-Vehicle PDPTW II

Algorithms for the single-vehicle PDTSP (no time windows). Without time windows, the problem simplifies to the PDTSP. Hernandez-Perez and Salazar-Gonzalez proposed a branch-and-cut that uses a clever reformulation: they observe that the PDTSP is equivalent to the TSP on a reduced graph where infeasible arcs (those that would violate precedence or capacity) are removed. Their algorithm solves instances with up to 40 nodes.

Key Takeaway: Single-Vehicle PDPTW

The single-vehicle PDPTW is important as a sub-problem in column generation for the multi-vehicle case. Efficient algorithms for it (particularly those that generate optimal routes quickly) are critical to the scalability of branch-and-price methods for the full multi-vehicle PDPTW.

6.4.5 Multiple-Vehicle PDPTW I

In the multi-vehicle setting, a fleet K of identical vehicles must collectively serve all n requests. This is the standard industrial setting and the most studied variant in the literature.

6.4.5 Multiple-Vehicle PDPTW II

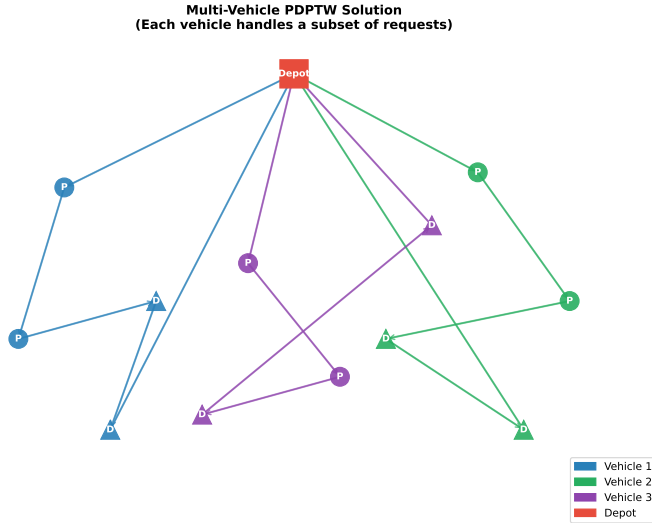


Figure: A multi-vehicle PDPTW solution with 3 vehicles (blue, green, purple). Each vehicle serves a subset of $_{40/79}$

Python: PDPTW Feasibility Check and Greedy Construction

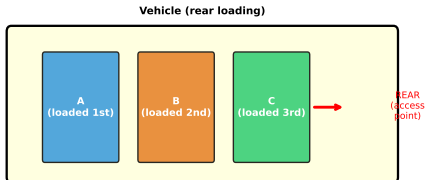
```
1  """
2  PDPTW feasibility checker and greedy insertion heuristic.
3  Assumes symmetric travel cost/time: dist[i][j] == dist[j][i].
4  """
5  from dataclasses import dataclass, field
6  from typing import List, Tuple, Optional
7  import math
8
9  @dataclass
10 class Request:
11     idx: int          # request id (1..n)
12     pickup: int       # pickup node index
13     delivery: int     # delivery node index
14     load: float       # goods quantity
15     e_p: float        # earliest service at pickup
16     l_p: float        # latest service at pickup
17     e_d: float        # earliest service at delivery
18     l_d: float        # latest service at delivery
19
20 def is_feasible(route: List[int], requests: dict, dist: List[List[float]],
21               capacity: float, service_time: float = 0.0) -> bool:
22     """
23     Check if a route is feasible for PDPTW.
24     route: list of node indices including depot (0) at start and end.
25     requests: dict mapping pickup_node -> Request object.
26     Returns True if precedence, capacity, and time windows are all satisfied.
27     """
```

6.5 Problems with Loading Constraints I

In practice, goods transported by vehicles are physical three-dimensional objects. The order in which goods are loaded onto a vehicle determines whether they can be unloaded at each delivery stop without disturbing other cargo. This gives rise to **loading-constrained PDPs** – problems in which both the routing and the loading plan must be co-optimised.

Loading Constraints: LIFO vs FIFO

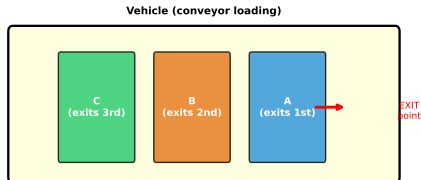
LIFO (Last-In First-Out)
- rear loading, stack policy



Must unload C, then B, then A (stack order)

LIFO: delivery sequence = reverse of loading sequence

FIFO (First-In First-Out)
- side/conveyor loading



A exits first (loaded first), then B, then C

FIFO: delivery sequence = same as loading sequence

6.5 Problems with Loading Constraints II

Figure: Left: LIFO loading – items loaded first (deepest inside the vehicle) must be delivered last; items near the rear are delivered first. Right: FIFO loading – items loaded first exit first (conveyor-style). The delivery sequence on the route must respect the physical loading order.

6.5 Problems with Loading Constraints III

Three types of loading constraints:

- ① **LIFO (Last-In First-Out):** the vehicle is accessed from the rear only (like a stack). The most recently loaded item must be unloaded first. This means the *delivery sequence must be the reverse of the loading sequence*: if item *A* is loaded before item *B*, then *B* must be delivered before *A*.
- ② **FIFO (First-In First-Out):** the vehicle is a conveyor – items loaded first exit first. The delivery sequence must *match* the loading sequence.
- ③ **3D Loading:** items have explicit three-dimensional extents (length, width, height) and must physically fit inside the container at every stage of the route. This requires solving a bin-packing sub-problem at each delivery stop.

6.5 Problems with Loading Constraints IV

LIFO Constraint: Formal Statement

Let σ be the route sequence of customers visited by a vehicle, and let π be the loading order (order in which items are placed onto the vehicle). The LIFO constraint states:

For any two items i and j carried by the same vehicle, if i is delivered before j (i.e., i comes before j in the route), then j must have been loaded before i (i.e., j precedes i in the loading order).

Equivalently: $\text{position}(\text{deliver } i) < \text{position}(\text{deliver } j) \Rightarrow \text{position}(\text{load } j) < \text{position}(\text{load } i)$.

6.5 Problems with Loading Constraints V

PDPTW with LIFO constraints. Carrabs, Cordeau, and Laporte (2007) studied the **PDPTW-L** (PDPTW with LIFO constraints) and showed that:

- LIFO constraints can be enforced without explicitly tracking the loading order: it suffices to check that the route does not contain any *interleaving* pair. Two requests i and j are **interleaved** if the route visits them in the order p_i, p_j, d_i, d_j (or p_j, p_i, d_j, d_i). LIFO feasibility requires that no such interleaving occurs.
- Equivalently: for every pair of requests on the same vehicle, the pickup-delivery intervals must be either *nested* (one contained within the other) or *disjoint* (non-overlapping).

6.5 Problems with Loading Constraints VI

LIFO Interleaving: Worked Example

Suppose a route visits nodes in order: Depot, P_1 , P_2 , D_1 , D_2 , Depot.

- Requests: (P_1, D_1) and (P_2, D_2) .
- Position of $P_1 = 1$, $P_2 = 2$, $D_1 = 3$, $D_2 = 4$.
- The sequence $P_1 < P_2 < D_1 < D_2$ means the intervals $[1, 3]$ and $[2, 4]$ *overlap but neither is contained in the other* – this is **interleaved**.
- A LIFO-feasible alternative: Depot, P_1 , P_2 , D_2 , D_1 , Depot gives intervals $[1, 4]$ and $[2, 3]$ – interval $[2, 3]$ is **nested** inside $[1, 4]$. This satisfies LIFO: item 2 (loaded after item 1 at P_2) is delivered first at D_2 , then item 1 at D_1 . ✓

6.5 Problems with Loading Constraints VII

Algorithms for PDPTW-L. Carrabs et al. (2007) designed a Branch-and-Cut algorithm for PDPTW-L that incorporates specific valid inequalities for LIFO feasibility. The branch-and-cut was able to solve instances with up to 25 requests.

For larger instances, heuristic approaches are necessary:

- **Tabu Search:** the neighbourhood moves must be restricted to those that maintain LIFO feasibility. A swap move that creates an interleaving is rejected.
- **LNS with LIFO repair:** after removing a subset of requests, the repair heuristic inserts them at positions that maintain the nesting property.
- **Dynamic programming:** for a single vehicle, DP can track the set of currently open (picked up but not yet delivered) requests and ensure LIFO ordering at each step.

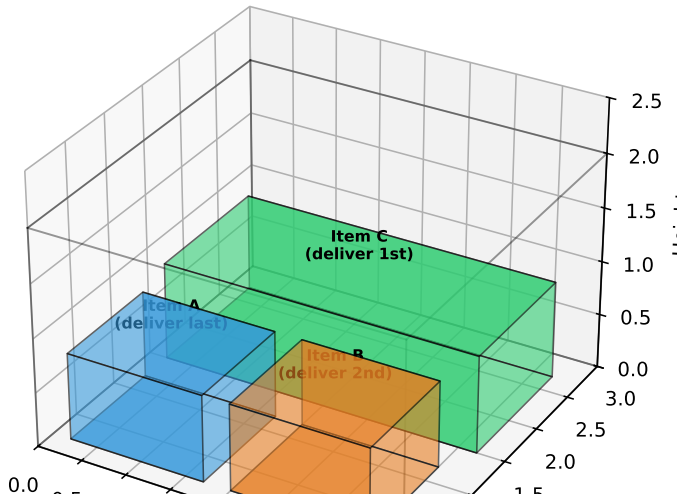
PDPTW with FIFO constraints. The PDPTW-F (FIFO variant) has analogous structure: instead of nesting, the pickup-delivery intervals must be *compatible* in the FIFO sense. Ladier and Alpan (2016) and others have shown that FIFO constraints arise naturally in conveyor-loading contexts such as airport baggage handling.

6.5 Problems with Loading Constraints VIII

3D Loading PDP. The most complex loading variant requires that items be physically packed into the vehicle's container in three dimensions. At every delivery stop, the item to be unloaded must be accessible from the door without requiring other items to be moved.

6.5 Problems with Loading Constraints IX

3D Loading Constraints in PDP
(Items must fit and be accessible at each delivery stop)



6.5.1 One-to-One Problems with Loading Constraints I

The one-to-one structure is the most natural setting for loading-constrained PDPs, since each request has a dedicated item to be picked up and delivered. Research has focused on two specific settings:

Single-vehicle PDPTW with LIFO. Carrabs, Cordeau, and Laporte (2007) proposed a Branch-and-Cut based on the following key result: LIFO feasibility of a route can be verified in $O(n)$ time by scanning the route and maintaining a stack. If, when attempting to deliver item i , the top of the stack is not i , the route is LIFO-infeasible.

Algorithm (LIFO Feasibility Check):

- ① Initialise an empty stack S .
- ② Scan the route from the depot to the end:
 - If the current node is a **pickup** p_i : push i onto S .
 - If the current node is a **delivery** d_i : if $\text{top}(S) = i$, pop S ; else the route is **LIFO-infeasible**.
- ③ If the stack is empty at the end, the route is LIFO-feasible.

6.5.1 One-to-One Problems with Loading Constraints II

LIFO Stack Check: Example

Route: Depot, P_1 , P_3 , D_3 , P_2 , D_2 , D_1 , Depot.

Node	Action	Stack
P_1	Push 1	$\{1\}$
P_3	Push 3	$\{1, 3\}$
D_3	Top = 3, Pop	$\{1\}$
P_2	Push 2	$\{1, 2\}$
D_2	Top = 2, Pop	$\{1\}$
D_1	Top = 1, Pop	$\{\}$

Stack empty at end: route is **LIFO-feasible**. ✓ The structure shows nested intervals: $[P_1, D_1]$ contains $[P_2, D_2]$, which is disjoint from $[P_3, D_3]$.

6.5.1 One-to-One Problems with Loading Constraints III

One-to-Many-to-One with Loading Constraints. In the 1-M-1 setting, Salazar-Gonzalez (2014) considered a variant where the same vehicle delivers goods from the depot (forward logistics) and collects returned goods (reverse logistics) on the same tour. LIFO constraints arise because the vehicle is loaded at the depot in a specific order that constrains the delivery sequence.

Incorporating loading into the MIP. In principle, loading constraints can be incorporated directly into the integer programme by adding binary variables $l_{ij} \in \{0, 1\}$ indicating whether item i is loaded before item j , with constraints linking l_{ij} to the route variables x . However, this leads to a significantly larger MIP. The preference in practice is to use the stack-based feasibility check as an efficient oracle within a metaheuristic.

Key Insight: LIFO as a Stack

The key algorithmic insight for LIFO-constrained PDPs is that LIFO feasibility is equivalent to the route forming a valid parenthesisation when pickup nodes are treated as opening parentheses and delivery nodes as closing parentheses. This allows $O(n)$ feasibility checking and simple neighbourhood filtering in TS and LNS.

6.6 Conclusions and Future Research Directions I

This chapter has presented the main variants of Pickup-and-Delivery Problems for goods transportation, classified by the relationship between origins and destinations (M-M, 1-M-1, 1-1) and by additional constraints (time windows, loading).

Summary of major algorithms:

Problem	Best Exact	Best Heuristic	Max Size
1-PDTSP (M-M, 1 vehicle)	B&C (H-P & S-G)	VNS, GRASP	200 nodes
M-PDTSP (M-M, multi-veh)	B&C	GRASP+VND (Qu&Bard)	300 nodes
VRPSPD (1-M-1)	B&C	Tabu Search, LNS	200 cust.
PDPTW (1-1, multi-veh)	B&C (Cordeau)	ALNS (Ropke&Pisinger)	1000 req.
PDPTW-L (1-1 + LIFO)	B&C	TS, LNS	100 req.
3L-PDPTW (3D loading)	—	2-level meta.	50 req.

Dominant research themes:

6.6 Conclusions and Future Research Directions II

- ① **Exact methods** have matured significantly, with branch-and-cut capable of solving mid-size PDPTW instances (up to 75 requests) in reasonable time. Column generation and branch-and-price extend this somewhat further.
- ② **ALNS and its variants** have become the standard metaheuristic for PDPs, largely displacing Tabu Search for single-objective problems. The key insight is the adaptive operator selection mechanism.
- ③ **Loading constraints** (LIFO, FIFO, 3D) have received increasing attention as the practical importance of physical loading feasibility is recognised. The coupling of routing and packing sub-problems remains a significant challenge.

6.6 Conclusions and Future Research Directions III

Open research directions:

- 1 **Dynamic PDPs:** in many real applications (courier delivery, ride-hailing), requests arrive in real time. Reoptimisation strategies that update the routing plan as new requests arrive, with minimal disruption to ongoing routes, remain an active research frontier.
- 2 **Stochastic PDPs:** demand quantities or travel times may be uncertain. Robust and stochastic optimisation formulations for PDPs are less mature than for standard VRPs.
- 3 **Multi-commodity PDPs:** when different items must not be mixed (e.g., food-grade and non-food-grade goods), the loading constraints become a multi-commodity assignment problem.
- 4 **Green PDPs:** minimising fuel consumption or CO₂ emissions rather than distance leads to different optimal routes, especially when vehicle weight (and thus fuel consumption) changes as goods are picked up and delivered.
- 5 **Crowd-sourced delivery:** platforms that use ordinary citizens as carriers (as in Uber Freight) create large-scale PDPs with hundreds of thousands of requests and require fundamentally new algorithmic approaches.

6.6 Conclusions and Future Research Directions IV

- 6 **Integration with inventory management:** joint optimisation of routing and inventory replenishment decisions leads to PDPs with vendor-managed inventory (VMI) structure, where the carrier also decides delivery quantities.

Chapter Summary

Pickup-and-Delivery Problems for goods transportation are a rich and practically important family of combinatorial optimisation problems. The PDPTW has emerged as the canonical problem, with ALNS as the dominant metaheuristic and branch-and-cut as the leading exact method. Loading constraints (LIFO, FIFO, 3D) add physical realism at the cost of significantly increased algorithmic complexity. Future research will increasingly focus on dynamic, stochastic, and large-scale variants driven by the growth of e-commerce and sharing-economy platforms.

Python: Large Neighbourhood Search (LNS) for PDPTW

```
1 """
2 Simplified LNS (Large Neighbourhood Search) for PDPTW.
3 Uses Shaw removal (relatedness-based) and greedy insertion repair.
4 Accepts worsening moves with simulated annealing probability.
5 """
6 import math, random, copy
7
8 def shaw_remove(routes, requests_info, dist, q=3):
9     """
10     Remove q related requests from solution.
11     Relatedness: geometric proximity (close pickup-to-pickup distance).
12     Returns (new_routes, removed_list_of_request_ids).
13     """
14     # Flatten all served requests: list of (req_id, route_idx, p_pos, d_pos)
15     served = []
16     for r_idx, route in enumerate(routes):
17         for pos, node in enumerate(route):
18             if node in requests_info and requests_info[node]['type'] == 'pickup':
19                 req_id = requests_info[node]['req_id']
20                 served.append((req_id, r_idx))
21     if not served:
22         return routes, []
23     # Pick a random seed request
24     seed_req, _ = random.choice(served)
25     removed = [seed_req]
26     # Remove q-1 most similar requests (smallest pickup-to-pickup distance)
27     seed_p = requests_info[seed_req + 100]['pickup'] # pickup node
```

Price Method and OR-opt for the 1-M-1 PDP I

For the One-to-Many-to-One PDP, one of the earliest systematic approaches was the **Price method**, introduced by Mosheiov (1994) for the case of deliveries followed by pickups (the backhauls variant). The idea is to decompose the problem into a delivery tour and a pickup tour and to combine them cost-effectively.

OR-opt Move for VRPSPD

An **OR-opt** move takes a customer (or a chain of k consecutive customers) from its current position in a route and reinserts it at another position in the same or a different route. For the VRPSPD, the move must be feasibility-checked for:

- 1 Capacity: at every arc in the new route, the combined delivery and pickup load must not exceed Q .
- 2 No time-window violation if time windows are imposed.

OR-opt(1) considers single-customer relocations; OR-opt(2) and OR-opt(3) consider chains of 2 or 3 consecutive customers. Combined, these give a rich neighbourhood of manageable size $O(n^2)$ per operator.

Montane and Galvao (2006) Tabu Search for VRPSPD. The algorithm maintains a tabu list of (*customer, removed-from-route*) pairs. A move is tabu if the customer was recently removed from a given route; it is accepted despite the tabu status only if the new solution beats the best solution found so far (aspiration criterion). The authors tested on 14 benchmark instances (16 to 100 customers) and reported improvements over the previous best solutions in 11 out of 14 cases.

Price Method and OR-opt for the 1-M-1 PDP III

Worked Example: OR-opt(1) for VRPSPD with $Q = 8$

Current solution:

- Route 1: Depot $\rightarrow$ C1($d = 3, p = 1$) $\rightarrow$ C3($d = 2, p = 2$) $\rightarrow$ Depot. Loads: $[6, 3 + 1, 1 + 2, 0 + 2] = [6, 4, 3, 2]$. All ≤ 8 . ✓
- Route 2: Depot $\rightarrow$ C2($d = 1, p = 4$) $\rightarrow$ C4($d = 2, p = 1$) $\rightarrow$ Depot. Loads: $[3, 3 - 1 + 4, 2 + 4 - 2 + 1, 0 + 1] = [3, 6, 5, 1]$. All ≤ 8 . ✓

OR-opt(1): Move C4 from Route 2 to Route 1 between C1 and C3.

- Route 1 new: Depot $\rightarrow$ C1 $\rightarrow$ C4($d = 2, p = 1$) $\rightarrow$ C3 $\rightarrow$ Depot. Loads: $[6 + 3 - 2 + 2, 4 - 2 + 1, 3 + 1 - 2, 0 + 2] = [9, 3, 2, 2]$. Load $9 > 8$. **Infeasible – move rejected.**

The OR-opt move is rejected because the initial load including C4's delivery would exceed capacity. The feasibility check prevents the route from becoming infeasible.

Price Method and OR-opt for the 1-M-1 PDP IV

Bianchessi and Righini (2007) heuristics. These authors proposed four heuristics for the VRPSPD, including a **greedy randomised construction** combined with **2-opt*** improvement (edge exchange between two different routes):

2-opt* Neighbourhood for VRPSPD

A **2-opt*** move considers two routes r_1 and r_2 , selects a *cut point* in each, and reconnects the tails: the tail of r_1 after the cut is appended to the head of r_2 up to its cut point, and vice versa. This exchanges large segments between routes and can dramatically improve the solution when two routes share a geographic region. For VRPSPD, feasibility of both new routes must be verified after the exchange.

Key Takeaway: 1-M-1 Heuristics

The VRPSPD is best approached with metaheuristics that combine OR-opt and 2-opt* moves with a flexible acceptance mechanism (tabu list or simulated annealing). The key challenge is feasibility checking: unlike standard VRP, the load on every arc (not just in total) must be checked after each move, which multiplies the cost of neighbourhood evaluation by a factor of $O(n)$.

Branch-and-Price for Multi-Vehicle PDPTW I

Set-partitioning formulation. The multi-vehicle PDPTW can be reformulated as a set-partitioning problem in which each column represents a complete, feasible route. Let Ω be the set of all feasible routes, and let c_r be the cost of route $r \in \Omega$. Then:

Set-Partitioning Formulation (SP)

$$\min_{\lambda} \quad \sum_{r \in \Omega} c_r \lambda_r \quad (11)$$

$$\text{s.t.} \quad \sum_{r \in \Omega} a_{ir} \lambda_r = 1 \quad \forall i \in \{1, \dots, n\} \quad (12)$$

$$\lambda_r \in \{0, 1\} \quad \forall r \in \Omega \quad (13)$$

Here $\lambda_r = 1$ indicates that route r is selected, and $a_{ir} = 1$ if route r serves request i . Each request must appear in exactly one selected route.

Branch-and-Price for Multi-Vehicle PDPTW II

Because the number of feasible routes $|\Omega|$ is exponential, the LP relaxation is solved by **column generation**: start with a restricted master problem (RMP) containing a small set of routes, then iteratively add routes with negative reduced cost.

Pricing sub-problem. Adding a route with the most negative reduced cost requires solving a **Shortest Path Problem with Resource Constraints (SPPRC)**: find a path from the origin depot to the destination depot that minimises the reduced cost subject to all PDPTW constraints (precedence, capacity, time windows). This is solved by dynamic programming on the time-expanded graph.

Branch-and-Price for Multi-Vehicle PDPTW III

Branch-and-Price algorithm.

- 1 **Root node:** solve LP relaxation of the set-partitioning formulation via column generation. If optimal LP solution is integer, stop.
- 2 **Branching:** if the LP solution is fractional, branch on a variable. A common branching strategy for VRPs is to branch on arc flows: force an arc (i,j) to be used (or not) in the selected routes.
- 3 **At each node:** resolve the column generation problem with the branching constraint incorporated into the pricing sub-problem.
- 4 **Pruning:** prune nodes where the LP lower bound exceeds the incumbent.

Key difficulty. The SPPRC pricing problem for the PDPTW is NP-hard itself (because of the precedence constraints), which makes each column generation iteration expensive. In practice, *heuristic pricing* (finding columns with negative reduced cost but not necessarily the most negative) is used to speed up convergence.

Results. Dumas, Desrosiers, and Soumis (1991) first applied column generation to the PDPTW, solving instances with up to 35 requests. Subsequent improvements by Ropke et al. (2007) extended the range to 75 requests using stronger branching and cutting-plane integration.

Key Insight: Column Generation Duality

The dual variables of the set-partitioning LP have a clear economic interpretation: π_i is the “shadow price” of serving request i , meaning the marginal cost reduction from removing the constraint that request i must be served. A route is worth adding to the master problem if and only if its travel cost minus the sum of shadow prices of the requests it serves is negative (negative reduced cost).

Benchmark Instances and Computational Results I

Li and Lim (2003) benchmark for PDPTW. This is the standard test set for the multi-vehicle PDPTW. It consists of instances derived from Solomon's VRPTW benchmarks by generating pickup-delivery pairs. Six classes are defined:

Class	Customer geometry	Time window type
LC1	Clustered	Tight
LC2	Clustered	Wide
LR1	Random	Tight
LR2	Random	Wide
LRC1	Mixed (random+clustered)	Tight
LRC2	Mixed (random+clustered)	Wide

Each class contains 12 instances with 100 requests (200 nodes plus depot copies). Wide time-window instances (LC2, LR2, LRC2) generally allow more flexible routing and tend to yield fewer vehicles but longer routes.

Benchmark Instances and Computational Results II

Key algorithmic results on Li-Lim benchmarks.

Algorithm	Authors	Year	Avg. gap
Tabu Search	Li & Lim	2003	$\approx 3\%$
LNS	Ropke & Pisinger	2006	$\approx 1\%$
ALNS	Ropke & Pisinger	2006	$< 1\%$
Branch-and-Cut (exact)	Cordeau	2006	0% (optimal)
B&C + LNS (matheuristic)	Ropke et al.	2007	$< 0.5\%$
POPMUSIC	Taillard et al.	2013	$< 0.3\%$

Instance sizes solved to optimality. The exact branch-and-cut of Cordeau (2006) solves instances with up to 75 requests within a 2-hour time limit. For instances with 100 requests, optimal solutions are known for some but not all problem instances.

Li-Lim benchmark notation. The instances are often referred to by codes such as “lc101” (Li-Lim, clustered, tight windows, instance 1), “lr211” (Li-Lim, random, wide windows, instance 11), and so on. The benchmark results are maintained online and updated as new best-known solutions are found by the research community.

Benchmark Instances and Computational Results III

Benchmarks for loading-constrained PDPs.

- **PDPTW-L (LIFO):** Carrabs, Cordeau, and Laporte (2007) used a set of randomly generated instances with up to 25 requests for exact methods, and up to 100 requests for heuristic evaluation.
- **VRPSPD:** Montane and Galvao (2006) used the instances of Dethloff (2001) and Angelelli and Mansini (2002), with 16 to 200 customers. These are derived from standard CVRP instances by adding a random pickup demand.
- **3L-PDPTW:** No standardised benchmark exists; researchers typically generate instances using standard routing instances (Solomon, Li-Lim) and add box dimensions from 3D packing literature.

Key Takeaway: Benchmarks and Progress

The Li-Lim benchmark has driven significant progress in PDPTW algorithms over two decades. ALNS (Ropke & Pisinger, 2006) was a landmark result that established LNS as the dominant metaheuristic paradigm. The benchmark remains partially open: for large instances (200+ requests) the optimality of known solutions is unconfirmed.

Genetic Algorithms for the PDPTW I

Genetic algorithms (GAs) have been applied to the PDPTW by several research groups. The central challenge is *chromosome encoding*: the standard order-based encoding used for the TSP must be adapted to respect both the fleet structure (multiple vehicles) and the precedence constraint (pickup before delivery on the same vehicle).

Chromosome Encoding for PDPTW

One effective encoding uses a **sequence of request IDs** (not node IDs):

- 1 A chromosome $C = (r_1, r_2, \dots, r_n)$ is a permutation of the n request IDs.
- 2 A **decoding procedure** converts this permutation into routes: it scans C from left to right and assigns each request to the current vehicle's route. When inserting request r_k , the pickup is placed before the delivery, maintaining precedence automatically.
- 3 When the route would violate capacity or time windows for the next request, a new vehicle route is opened.

This encoding guarantees that the precedence constraint is always satisfied by construction.

Crossover operators. The standard **Order Crossover (OX)** is adapted as follows:

Genetic Algorithms for the PDPTW II

- ① Select a subsequence from parent P_1 .
- ② Copy this subsequence directly into the offspring.
- ③ Fill the remaining positions with the requests from P_2 in the order they appear, skipping requests already in the offspring.
- ④ Decode the offspring chromosome into routes using the decoding procedure.

Genetic Algorithms for the PDPTW III

Mutation operators. Common mutation operators for PDPTW GAs:

- **Pair swap:** swap two randomly chosen request IDs in the chromosome.
- **Subsequence reversal:** reverse a randomly chosen subsequence of request IDs.
- **Request relocation:** remove a request ID from its position and reinsert it at a random position later in the chromosome (maintaining the chromosome property that pickup precedes delivery by construction).

Population diversity. A key challenge in GA for PDPTW is maintaining population diversity. Without diversity mechanisms, the population quickly converges to a single region of the solution space (genetic drift). Common remedies:

- **Crowding replacement:** new offspring replace the most similar existing chromosome rather than the worst.
- **Island model:** maintain multiple sub-populations that evolve independently and occasionally exchange chromosomes (migration).
- **Fitness sharing:** penalise solutions that are too similar to other solutions already in the population.

Genetic Algorithms for the PDPTW IV

Hybrid GA. Vidal et al. (2013) proposed a highly effective **Hybrid Genetic Search (HGS)** that integrates:

- ① A GA for global diversification (population-level exploration).
- ② A local search procedure (a powerful sequence of neighbourhood moves) applied to every offspring before it is added to the population.
- ③ A survivor selection that balances fitness with diversity.

HGS has produced state-of-the-art results on multiple VRP variants, including the PDPTW.

Key Insight: Why Hybridise?

Pure GAs struggle with PDPTW because crossover operators tend to produce infeasible or low-quality offspring that require expensive repair. Hybridising with local search (creating a “memetic algorithm”) ensures that every individual in the population is a locally optimal solution, dramatically improving convergence speed.

Chapter Summary: Algorithm Landscape I

When to use which algorithm? The choice depends on three factors: (1) problem variant (M-M, 1-M-1, 1-1, with/without loading), (2) instance size, and (3) available computation time.

Scenario	Exact method	Metaheuristic	Scalability
M-M, 1 vehicle ($n \leq 40$)	Branch-and-Cut (Hernandez-Perez)	VNS, GRASP	Up to 200 nodes
M-M, multi-vehicle	Branch-and-Cut	GRASP+VND (Qu & Bard)	Up to 300 nodes
1-M-1 / VRPSPD ($n \leq 25$)	Branch-and-Cut	Tabu Search, LNS	Up to 500 customers
PDPTW ($n \leq 75$)	Branch-and-Cut (Cordeau)	ALNS (Ropke & Pisinger)	Up to 1000 requests
PDPTW with LIFO ($n \leq 25$)	Branch-and-Cut (Carrabs)	LNS with stack repair	Up to 200 requests
3D-Loading PDPTW	No exact method	2-level meta (route + pack)	Up to 50 requests

Chapter Summary: Algorithm Landscape II

The dominance of ALNS. Across all 1-1 PDP variants, ALNS (Ropke & Pisinger, 2006) has emerged as the most versatile and effective algorithm. Its key advantages are:

- ➊ **Modularity:** new destroy/repair operators can be added without changing the core algorithm. This makes it easy to extend ALNS to new problem variants.
- ➋ **Automatic tuning:** the adaptive weight update mechanism reduces the need for manual operator tuning – the algorithm self-adjusts to the instance structure.
- ➌ **Scalability:** unlike exact methods (exponential) or Tabu Search (quadratic neighbourhood), LNS explores large moves in $O(qn)$ time, allowing it to handle instances with thousands of requests.
- ➍ **Quality:** on the Li-Lim benchmarks, ALNS achieves solutions within 1% of optimal for most instances, comparable to branch-and-cut on instances where exact methods are feasible.

Research trajectory. The field has moved from:

- **1990s:** exact methods (B&C, column generation) for small instances.
- **2000s:** construction heuristics and Tabu Search for medium instances.
- **2006–present:** LNS/ALNS dominance for large instances; exact methods strengthened by better cutting planes and branch-and-price.

Chapter Summary: Algorithm Landscape III

- **Emerging:** machine learning-guided heuristics, dynamic PDPs, and integration with real-time data (GPS, demand forecasting).

Final Takeaway

The Pickup-and-Delivery Problem with Time Windows (PDPTW) is the canonical hard problem in goods transportation logistics. It combines the difficulty of the Travelling Salesman Problem (routing) with resource constraints (capacity, time windows) and coupling constraints (precedence, pairing). ALNS – a meta-framework that combines large-scale destroy-and-repair moves with adaptive operator selection – currently provides the best practical trade-off between solution quality and computational cost.

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